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Analysis of Feature Importance in a Neural Network for Predicting the Remaining Useful Life of Aircraft Engines

Master's Thesis
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List of Symbols

Symbol	Unit	Description
Latin symbols		
F_{xy}		Faktor xy
H	m	Höhe
c_p	$\mathrm{J/(kg\,K)}$	Spezifische Wärmekapazität
Greek symbols		
κ		Isentropenexponent
λ	$\mathrm{W/(m\,K)}$	Wärmeleitfähigkeit
ε	$\mathrm{m^2/s^3}$	Dissipationsrate
Similarity parameters		
Bi		Biotzahl
Nu		Nußeltzahl
Pr		Prandtlzahl
Re		Reynoldszahl
Indices		
r		r-Koordinate
x		x-Koordinate
y		y-Koordinate
Abbreviations		
CFD		Computational Fluid Dynamics
FEM		Finite-Elemente-Methode

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1 Introduction

The aviation industry is one of the most safety-critical and technologically sophisticated fields in the world. Ensuring the reliability and availability of aircraft is essential not only for passenger safety but also for the economic efficiency of airlines and engine manufacturers. As engines operate under extreme mechanical and thermal conditions, their degradation processes must be continuously monitored and understood. In recent years, predictive maintenance has become a central concept in aviation, aiming to anticipate component failures before they occur and to enable maintenance actions precisely when needed.

A key element of predictive maintenance is the estimation of the Remaining Useful Life (RUL) of engine components. RUL predictions allow operators to plan maintenance more effectively, reduce unscheduled downtime, and avoid unnecessary part replacements. However, obtaining accurate RUL predictions is challenging. The underlying degradation mechanisms are highly nonlinear and influenced by a large number of operational parameters. Traditional modelling techniques often struggle to capture this complexity.

With the rise of large sensor datasets and computational power, machine learning and deep learning methods have become increasingly important in RUL estimation. Neural networks, in particular, are capable of modelling complex relationships in engine data and have demonstrated strong predictive performance. However, these models introduce a new challenge: despite their accuracy, they often behave as black boxes, offering little insight into how individual features contribute to the prediction. In a safety-critical environment such as aviation, this lack of transparency poses a significant limitation. Understanding why a model arrives at a certain prediction is essential for trust, certification, and operational decision-making.

This challenge has led to the emergence of Explainable Artificial Intelligence (XAI). XAI methods aim to make complex AI systems more interpretable by providing insights into which input feature influence a model's output and to what extent. One of the most widely used approaches within XAI is the analysis of feature importance. Feature importance methods allow engineers and researchers to understand which parameters are most relevant for RUL estimation and how different models may focus on different aspects of the data.

2 Related Work

2.1 Data-Driven Remaining Useful Life Prediction

The dataset used throughout this tesis is C-MAPSS (Commercial Modular Aero-Propulsion System Simulation), published by Saxena et al. (2008) for the PHM 2008 data challenge. It contains four subsets of simulated turbofan engine run-to-failure trajectories, FD001 through FD004, each with a different combination of operating conditions and fault modes. Every engine unit records 21 sensor measurements and three operatinoal settings per cycle. Degradation is injected into the high-pressure compresso(HPC), the fan, or both, depending on the subset. 2.1 gives the breakdown.

Subset	Training Engines	Test Engines	Operating Conditions	Fault Modes
FD001	100	100	1 (Sea Level)	1 (HPC Deg)
FD002	260	259	6	1 (HPC Deg)
FD003	100	100	1 (Sea Level)	2 (HPC + Fan Deg)
FD004	248	249	6	2 (HPC + Fan Deg)

Table 2.1: Overview of the C-MAPSS dataset subsets Saxena et al. (2008).

FD001 is the simples case, one operating condition, one fault mode. FD004, with six operating conditioins and two fault modes, is the hardest. The multi-condition subsets (FD002 and FD004) contain roughly two and a half times as many engines, reflecting the additional variablility. C-MAPSS has appeared in well over a hundred publications since its release and remains the go-to benchmakr, mostly because nothing else offers the same scale with controlled degradation across multiple regimes.

Nearly all C-MAPSS studies adopt a piecewise linear(PwL) labelling scheme for the RUL target. Heimes (2008) showed with a multilayer perceptron that prediction improve when the early, healthy portion of each trajectory is assigned a constant RUL rather than a linearly decreasing one. The reasoning is straightforward: sensors show no measurable degradation in the early cycles, so there is nothing in the data to distinguish an engine at cycle 10 from one at cycle 50. Assigning a constant cap avoids forcing the model to learn distinctions that the sensor readings do ot support. Heimes set this cap at 130 cycles. Later work has used values between 115 and 130; this thesis uses 125, which is among the most common choices. The value is heruistic, and it matters, particularly on FD002 and FD004, where trajectory lengths vary more.

The first generation of data-driven RUL methods relied on classical machine learning. Wang et al. (2008) won the PHM 2008 challenge with a similarity-based algorithm. Support vector machines, random forests and ensemble approaches followed. Ramasso and Saxena (2014) reviewed this body of work and noted a recurring dependence on hand-crafted health indicators, features built from domain knowledge about which sensors degrade and how. This is effective, but hard to scale.

Deep learning changed that. Sateesh Babu et al. (2016) were among the first to apply a deep Convolutional Neural Network (CNN) to RUL prediction, treating the problem as a regression over multivariate time-series inputs. Li et al. (2018) extended this direction with a deeper architecture and a systematic sliding-window sample preparation scheme, feeding raw sensor data directly into the network without manual feature engineering. Their model outperformed most prior methods on C-MAPSS and became one of the most cited CNN-based approaches in the prognostics literature. Long Short-Term Memory(LSTM)-based approaches appeared around the same time (Wu et al., 2018, Zheng et al., 2017), exploiting the sequential structure of the degradation data. Hybrid architectures combining CNN feature extraction with LSTM temporal modelling soon followed (e.g., Zhao et al. (2017)), and attention mechanisms were layered on top, for instance, Li et al. (2022) added a convolutional block attention module (CBAM) to a CNN-LSTM on C-MAPSS, allowing the model to weight sensor channels and time steps selectively. Transformer-based models and domain adaptation techniques are the most recent additions, through the core task, predicting a scalar RUL from a sensor window, has stayed the same throughout.

The model in this thesis is a 1D-CNN: convolutional layers for feature extraction, fully connected layers for regression. No recurrence, no attention, no gating. That structural simplicity matters here, because the focus is on interpretability. Fewer internal mechanism mean fewer confounding factors when attributing prediction back to individual input features.

2.2 Explaibale Artificial Intelligence in Prognostics

Strong RUL predictions are not enough on their own. In aviation maintenance, where sensor readings feed into decision about grounding aircraft or scheduling inspections, engineers need to understand what a model is responding to. A prediction driven by physically meaningful degradation signals is useful; one driven by spurious correlations in the training data is dangerous. This tension between predictive accuracy and interpretaility runs through the broader machine learning literature Rudin (2019), but it is especially acute in satey-critical prognostics, where a wrong explanatino attached to a correct prediction can mislead maintenance decisions just as badly as an incorrect prediction.

Applying XAI to prognostic models is still uncommon. Serradilla et al. (2020) applied Local Interpretable Model-agnostic Explanations (LIME; Ribeiro et al. (2016)) permutation-based feature importance to a random forest-based RUL estimator for industrial bushings, using the explanations alongside domain knowledge to guide model development. Theier work demonstrated the principle, that XAI tools can do more than just post-hoc reporting, but was limited to a relatively simple model on non-aviation data.

On C-MAPSS specifically, Hong et al. (2020) were among the first, using SHapley Additive exPlanations (SHAP; (Lundberg and Lee, 2017a)) on a deep CNN-Bidirectional LSTM. They first reduced the sensor set through correlation analysis and regularisation, then ran SHAP on the reduced inputs. The resulting force plots broke down individual predictions by sensor contribution. Useful as a proof of concept, but the analysis only covered one explanation method,

and because sensors had already been filtered before the SHAP step, the attributions described a pre-selected feature space rathen than what the model would have learned from the full suite.

Baptista et al. (2022) compared SHAP attributions against traditional prognostic metrics, monotonicity, trendability, prognostability, that are commonlgy used to assess sensor quality before modelling. The two did not always agree: sensor scoring well on prognostic metrics were not necessarily the ones the trained model weighted most heavily. That disconnect says something about the gap between a priori feature assessment and what a network actually extracts from the data during training.

Baptista et al. (2024) then turned to LIME (Ribeiro et al., 2016), applying it to a Gated Recurrent Unit(GRU) Network on C-MAPSS. LIME recovered both global and local feature importance, but the local explanation quality was uneven. Late in the degradation trajectory, where the signal is strong and clear, surrogate fidelity was reasonable. Early in life, with weak degradation signatures and noisy readings, it dropped off sharply, a known limitation of perturbation-based methods on time-series data, but worth confirming empirically.

Alomari et al. (2023) took SHAP in a different direction,, using it not just for post-hoc explanation but as part of the feature selection pipeline before training. Similar workflows have since appeared, coupling SHAP-based selection with CNN-LSTM or Temporal Convolutional Networks (TCN)-LSTM architectures. Gawde et al. (2024) applied four methods - SHAP, LIME, Partiall Dependence Plots (PDP), and Individual Conditional Expectation (ICE; Goldstein et al. (2015)) - to rotating machinery health monitoring, demonstrating a multi-method comparison, through on a different domain and with classical ML models rather than deep networks.

The methodological range across the C-MAPSS prognostics literature remains narrow. SHAP dominates. LIME appears occasionally. Gradient-based attribution methods, marginal-effect methods, and counterfactual approaches, all well estabileshed in the broader XAI field (see 3), are absent from C-MAPSS RUL work almost entirely. The few multi-method studies that do exist, like Gawde et al. (2024), apply to different equipment and simpler models. On deep networks trained for turbofan prognostics, no comparative evaluation has been carried out.

2.3 Research Gap and Thesis Positioning

The literature reviewed above reveals two things. First, deep learning models for RUL prediction on C-MAPSS have matured rapidly, from basic CNNs and LSTMs to attention-augmented hybrids and Transformers, but the question of what these models actually learn from the sensor data has received far less attention. Second, the small body of work that does address interpretability in this domain relies almost exlusively on SHAP, occasionally LIME, applied in isolation to a single architecture.

This is a problem, because different explanation methods operate on different principles and can produce differnt, sometimes contradictory, results for the same model and input. A gradient-based method like Integrated Gradients (Sundararajan et al., 2017) traces attributions through the network's own parameters. A perturbation-based method like Permutation Feature Impor-

tance (Breiman, 2001) observes how prediction change when inputs are shuffled. A marginal-effect method like Accumulated Local Effects (ALE; Apley and Zhu (2019)) characterise the prediction surface shifts as a single feature varies. Relying on one method alone risks mistaking that method's particular perspective for a complete picture of the model's behaviour.

This thesis addresses the gap by applying eight feature importance methods (see 3) to a 1D CNN trained on a variety of subset combinations of all four C-MAPSS subsets. The contribution is not just an expanded set of explanations, but a systematic comparison, identifying where methods agree on which sensor drive the model's prediction, where they diverge and what effect the different subset combinations have on the model.

3 Fundamentals

This chapter covers the theoretical foundations of the feature importance methods used in this thesis, including formal definitions, mathematical formulations, and key properties of each technique. The methods are divided into model-agnostic and model-specific approaches; application-specific details are addressed in Chapter 4.

3.1 Model-Agnostic Methods

Model-agnostic feature importance methods operate independently of the underlying model structure and require only the ability to query the model with input data and observe the resulting output. Let

$$f : \mathbb{R}^d \to \mathbb{R} \tag{3.1}$$

be a trained predictive model mapping an input vector $\mathbf{x} = (x_1, x_2, \ldots, x_d)$ to a scalar output such as a predicted RUL. This definition of f and $\mathbf{x}$ is retained throughout this section. Without access to the model internals, these methods analyse how the output changes when inputs are perturbed, fixed, permuted, or replaced, yielding either global or local explanations depending on whether effects are evaluated across the full dataset or at the level of individual samples.

3.1.1 Individual Conditional Expectation

Individual Conditional Expectation (ICE) visualises how a model's prediction changes as a single feature varies while all other features are held fixed at their observed values (Goldstein et al., 2015). For a sample $\mathbf{x}^{(i)} = (x_1^{(i)}, \ldots, x_d^{(i)})$, where superscript (i) identifies the sample and subscript j the feature of interest, the ICE curve is obtained by substituting a grid of values z into position j:

$$\mathrm{ICE}_j\left(\mathbf{x}^{(i)}, z\right) = f\left(x_1^{(i)}, \ldots, x_{j-1}^{(i)}, z, x_{j+1}^{(i)}, \ldots, x_d^{(i)}\right). \tag{3.2}$$

Each evaluation replaces x_j with z while all other features remain fixed, producing a prediction curve over the range of z for that instance. Averaging across all N samples gives the Partial Dependence Plot (PDP),

$$\mathrm{PDP}_j(z) = \frac{1}{N} \sum_{i=1}^{N} \mathrm{ICE}_j\left(\mathbf{x}^{(i)}, z\right), \tag{3.3}$$

which describes the average marginal effect of x_j across the data distribution. Where individual ICE curves diverge substantially, this signals feature interactions or non-linear effects that the PDP would obscure through averaging. In this thesis, a scalar importance score is derived from ICE by computing the mean variance of individual ICE curves across all instances,

$$\text{ICE-FI}_j = \frac{1}{M} \sum_{i=1}^{M} \mathrm{Var}_z\left[\mathrm{ICE}_j\left(\mathbf{x}^{(i)}, z\right)\right], \tag{3.4}$$

which measures how much each instance's prediction varies as feature j is swept across its range, averaged over all M instances (see Chapter 4). This captures instance-level sensitivity

more robustly than the range of the PDP, since it accounts for the full spread of the curve across all grid points rather than relying on the endpoints alone.

3.1.2 Feature Ablation

Feature Ablation quantifies feature importance by measuring how much the model's prediction changes when a feature is replaced by a neutral reference value (Zeiler and Fergus, 2014). If removing a feature's information causes a large shift in the output, that feature matters.

To evaluate feature x_j, a modified input $\tilde{\mathbf{x}}_j$ is constructed by replacing x_j with a reference value x_j^{ref}, such as the feature mean or zero, while all other features remain unchanged. The local attribution is

$$\phi_j(\mathbf{x}) = f(\mathbf{x}) - f(\tilde{\mathbf{x}}_j). \tag{3.5}$$

A large absolute value of $\phi_j(\mathbf{x})$ indicates strong local influence of feature j. In multivariate time-series data, entire sensor channels or time windows can be ablated jointly, enabling interpretation at a higher semantic level. Aggregating over the dataset gives a global importance score:

$$\phi_j^{\text{FA}} = \frac{1}{N} \sum_{i=1}^{N} \left| \phi_j\left(\mathbf{x}^{(i)}\right) \right|. \tag{3.6}$$

Attributions depend strongly on the choice of x_j^{ref}, and replacing features independently can generate inputs that lie far outside the training distribution when features are statistically dependent (Molnar, 2022).

3.1.3 Shapley Additive Explanations

SHAP (SHapley Additive exPlanations) attributes a model prediction to its input features using cooperative game theory (Lundberg and Lee, 2017b, Shapley, 1953). Each feature is treated as a player, and its attribution is its average marginal contribution to the prediction across all possible feature subsets.

A value function $v(S) \in \mathbb{R}$, defined for each subset $S \subseteq F = \{1, \ldots, d\}$, quantifies the model output when only the features in S are active, with absent features replaced by a reference value or marginalised out. The Shapley value for feature j is

$$\phi_j = \sum_{S \subseteq F \setminus \{j\}} \frac{|S|!\,(d - |S| - 1)!}{d!} \left(v(S \cup \{j\}) - v(S) \right), \tag{3.7}$$

where the factorial weight averages the marginal contribution $v(S \cup \{j\}) - v(S)$ over all coalition sizes and orderings (Shapley, 1953). The attributions form an additive decomposition of the prediction:

$$f(\mathbf{x}) = \phi_0 + \sum_{j=1}^{d} \phi_j, \tag{3.8}$$

where $\phi_0 = \mathbb{E}[f(X)]$ is a baseline value under a background distribution (Lundberg and Lee, 2017b). Four axioms uniquely characterise the Shapley values: *efficiency* ensures attributions sum to the explained quantity; *symmetry* gives equal attribution to features contributing identically to all coalitions; the *dummy property* assigns zero to features that never change the value function; and *linearity* ensures attributions combine linearly for linear combinations of models.

Two common formulations of $v(S)$ exist. Setting $v(S) = \mathbb{E}[f(X) \mid X_S = x_S]$ respects feature dependencies but requires estimating high-dimensional conditional distributions. Setting $v(S) = \mathbb{E}[f(\mathbf{x}_S, X_{\bar{S}})]$ draws absent features from a background distribution independently, which is computationally simpler but may evaluate the model at unlikely input combinations. Both formulations can yield different attributions when features are correlated (Aas et al., 2021). Exact computation requires evaluating all 2^d coalitions, which quickly becomes intractable; in practice, Shapley values are approximated via sampling-based estimation (Štrumbelj and Kononenko, 2014) or weighted regression (Lundberg and Lee, 2017b), at the cost of some variance in the estimates.

3.1.4 Local Interpretable Model-Agnostic Explanations

LIME (Local Interpretable Model-Agnostic Explanations) explains an individual prediction by fitting a simple interpretable surrogate to the original model in the neighbourhood of a specific input instance (Ribeiro et al., 2016). Even globally complex models often behave approximately linearly in a small enough region, which is the premise LIME exploits.

Given an instance $\mathbf{x} \in \mathbb{R}^d$, LIME samples perturbed inputs $\{\mathbf{z}_k\}_{k=1}^K$ near $\mathbf{x}$ and evaluates each to obtain $f(\mathbf{z}_k)$. A locality kernel $\pi_{\mathbf{x}}(\mathbf{z}_k)$ down-weights samples distant from $\mathbf{x}$, focusing the surrogate on the immediate neighbourhood of the instance. A sparse linear model g is then fitted by solving

$$\arg\min_{g \in \mathscr{G}} \mathscr{L}(f, g, \pi_{\mathbf{x}}) + \Omega(g), \tag{3.9}$$

where $\mathscr{L}$ measures how closely g approximates f under the locality weighting and $\Omega(g)$ penalises complexity (Ribeiro et al., 2016). For a linear surrogate,

$$g(\mathbf{z}) = w_0 + \sum_{j=1}^{d} w_j z_j, \tag{3.10}$$

the fitted coefficients w_j are the local feature importance scores, directly indicating each feature's contribution at $\mathbf{x}$. Explanations can vary across runs due to the stochastic nature of the perturbation sampling.

3.1.5 Counterfactual Explanations

Counterfactual explanations take a different perspective from attribution-based methods: instead of scoring each feature, they ask what the smallest change to the input would be to shift

the model's prediction to a desired target. Features that must change most are considered the most influential (Wachter et al., 2018).

For an instance $\mathbf{x} \in \mathbb{R}^d$ and a target y', a counterfactual $\mathbf{x}^{\text{cf}}$ is sought satisfying $f(\mathbf{x}^{\text{cf}}) = y'$ while minimising the distance $D(\mathbf{x}, \mathbf{x}^{\text{cf}})$. In practice this constrained problem is relaxed to

$$\mathbf{x}^{\text{cf}} = \arg\min_{\mathbf{z}} \lambda \cdot \mathscr{L}(f(\mathbf{z}), y') + D(\mathbf{z}, \mathbf{x}), \tag{3.11}$$

where $\mathscr{L}$ penalises deviation from y' and $\lambda > 0$ controls the trade-off between prediction fidelity and closeness to the original input (Wachter et al., 2018). D is typically an L_1 or L_2 norm. Additional constraints can be imposed to ensure plausibility, sparsity, or feasibility (Verma et al., 2022, **?**), and causal structure can be incorporated to respect feature dependencies (Kusner et al., 2018). The optimisation is generally non-convex, and multiple valid counterfactuals may exist for any given instance.

3.1.6 Permutation Feature Importance

Permutation Feature Importance (PFI) quantifies each feature's contribution by measuring the drop in predictive performance when the statistical link between that feature and the target is broken (Breiman, 2001, Fisher et al., 2019). Shuffling a feature's values across samples removes any information it carries about the target while leaving its marginal distribution unchanged, any resulting performance loss is then attributable to the loss of that feature's signal.

Let $\{(\mathbf{x}^{(i)}, y^{(i)})\}_{i=1}^{N}$ be the evaluation dataset and $\mathscr{L}(f)$ a performance metric such as mean squared error. For feature j, a permuted dataset $\tilde{\mathbf{x}}^{(j)}$ is constructed by randomly shuffling x_j across all samples while leaving all other features unchanged. The permutation importance is

$$\text{PFI}_j = \mathscr{L}\Big(f(\tilde{\mathbf{x}}^{(j)})\Big) - \mathscr{L}(f(\mathbf{x})), \tag{3.12}$$

where a larger value indicates greater model reliance on feature j for error-based metrics. PFI measures predictive dependence, not causal effect. When features are correlated, permuting one independently of the others creates combinations unlikely to appear in training, which can inflate or distort the estimates (Fisher et al., 2019).

3.1.7 Accumulated Local Effects

Accumulated Local Effects (ALE) estimates the average influence of a feature on model predictions in a way that accounts for feature correlations (Apley and Zhu, 2019). The standard alternative, PDP, averages predictions over the marginal distribution of all other features, which can produce evaluations at unrealistic input combinations when features are dependent. ALE addresses this by computing prediction differences within narrow local intervals, conditioning only on samples that actually fall in that region.

The domain of feature x_j is partitioned into K disjoint intervals at grid points $z_0 < z_1 < \cdots < z_K$, chosen as empirical quantiles to ensure sufficient data in each bin. For each sample $\mathbf{x}^{(i)}$ in

interval $[z_{k-1}, z_k]$, the local prediction difference is

$$\Delta_{j,k}^{(i)} = f\left(x_1^{(i)}, \ldots, z_k, \ldots, x_d^{(i)}\right) - f\left(x_1^{(i)}, \ldots, z_{k-1}, \ldots, x_d^{(i)}\right), \tag{3.13}$$

with all other features held at their observed values. The average within interval k,

$$\hat{\Delta}_{j,k} = \frac{1}{N_k} \sum_{i: x_j^{(i)} \in [z_{k-1}, z_k]} \Delta_{j,k}^{(i)}, \tag{3.14}$$

is then accumulated to give the ALE function at point z:

$$\mathrm{ALE}_j(z) = \sum_{k: z_k \leq z} \hat{\Delta}_{j,k}. \tag{3.15}$$

The function is centred so that

$$\int \mathrm{ALE}_j(z)\, dP_{X_j}(z) = 0 \tag{3.16}$$

meaning it reflects deviations from the average prediction rather than absolute values. Unlike ICE, which traces the full prediction curve per instance, ALE describes the average local rate of change across bins, a global perspective. The PDP formulation $\mathbb{E}[f(x_j, X_{\setminus j})]$, where $X_{\setminus j}$ denotes all features except j, averages over the marginal distribution, which can place the model in regions of the input space that are practically unreachable; ALE is specifically designed to avoid this. Instability in sparse regions of the feature space remains a limitation.

3.2 Model-Specific Methods

Model-specific feature importance methods make use of quantities internal to the model, gradients, activations, or parameter values and are therefore tied to a particular model class. Unlike model-agnostic methods, which work purely from input-output pairs, they require access to the model internals, restricting their applicability to differentiable architectures such as neural networks. Within that scope, they tend to be more computationally efficient and come with stronger theoretical guarantees.

3.2.1 Integrated Gradients

Integrated Gradients (IG) is a gradient-based attribution method for differentiable models (Sundararajan et al., 2017). Plain gradient saliency evaluates only the local gradient at the input, which can be uninformative when the model saturates. IG instead accumulates gradients along a straight-line path from a reference baseline to the instance being explained, building a more complete account of each feature's contribution.

Let $\mathbf{x} \in \mathbb{R}^d$ be the input and $\mathbf{x}' \in \mathbb{R}^d$ a baseline representing an uninformative reference state, for example, an all-zeros vector. The path between them is the vector-valued interpolation

$$\gamma(\alpha) = \mathbf{x}' + \alpha(\mathbf{x} - \mathbf{x}') \in \mathbb{R}^d, \quad \alpha \in [0, 1], \tag{3.17}$$

where α is a scalar interpolation parameter and $\gamma(\alpha)$ the corresponding input vector. The attribution for feature j is the path integral of the partial derivative of f with respect to x_j, scaled by the displacement of that feature between baseline and input:

$$\phi_j(\mathbf{x}) = (x_j - x'_j) \int_0^1 \frac{\partial f(\gamma(\alpha))}{\partial x_j} \, d\alpha. \tag{3.18}$$

In practice the integral is approximated by a Riemann sum with m steps:

$$\phi_j(\mathbf{x}) \approx (x_j - x'_j) \frac{1}{m} \sum_{k=1}^{m} \frac{\partial f(\gamma(\alpha_k))}{\partial x_j}, \quad \alpha_k = \frac{k}{m}. \tag{3.19}$$

IG satisfies the *completeness* property: attributions sum exactly to the difference in model output between input and baseline,

$$\sum_{j=1}^{d} \phi_j(\mathbf{x}) = f(\mathbf{x}) - f(\mathbf{x}'), \tag{3.20}$$

giving a fully additive decomposition of the prediction difference. Two further axioms are satisfied: *sensitivity*, a feature responsible for a prediction difference always receives non-zero attribution, and *implementation invariance*, functionally equivalent models produce identical attributions (Sundararajan et al., 2017). Attributions can differ substantially with different baseline choices, and no universally appropriate baseline exists. Global feature importance is obtained by aggregating local attributions across samples.

4 Methodology

This chapter describes the experimental setup used to analyse feature importance in neural network-based RUL prediction. It covers the dataset and preprocessing pipielin, the model arcchitecture and its three configurations, the training procedure, and the implementation of feature importance methods. The models themselves are not the primary contribution of this thesis; they serve as controlled predictive systems under which feature importance behaviour can be systematically studied across different input configurations.

4.1 Dataset and Data Preparation

4.1.1 Features set / sensors used

The raw input space of each CMAPSS sub-dataset consitst of three operational settings and 21 sensor measurements per time step. Eight sensors: s1, s5, s6, s10, s14, s16, s18, and s19, were identified as non-informative due to near-constant values across all trajectories and were excluded from the analysis. The remaining 13 sensor channels, listed in 4.1, form the feature set used throughout this work.

Table 4.1: Sensor features retained after non-informative sensor removal.

Senso ID	Physical Description
s2	LPC outlet temperature
s3	HPC outlet temperature
s4	LPT outlet temperature
s7	HPC outlet pressure
s8	Fan speed
s9	Core speed
s11	HPC outlet static pressure
s12	Ratio to Fuel flow to Ps30
s13	Corrected fan speed
s15	Bypass ratio
s17	Bleed enthalpy
s20	HPT cool air flow
s21	LPT cool air flow

Each sensor channel is scaled indepentently to the range [0, 1] using a MinMax scaler. Because sensosr readings vary systematically with ooperating condition, scaling is applied separately within each of the distinct operating regimes present in the data. The scaler is fitted exclusively on the training set.

Input sequences are constructed using a sliding window of length $T = 70$ time steps and stride 1, yielding tensors of shape (N x 30 x 70), where N is the number of sequences. The label

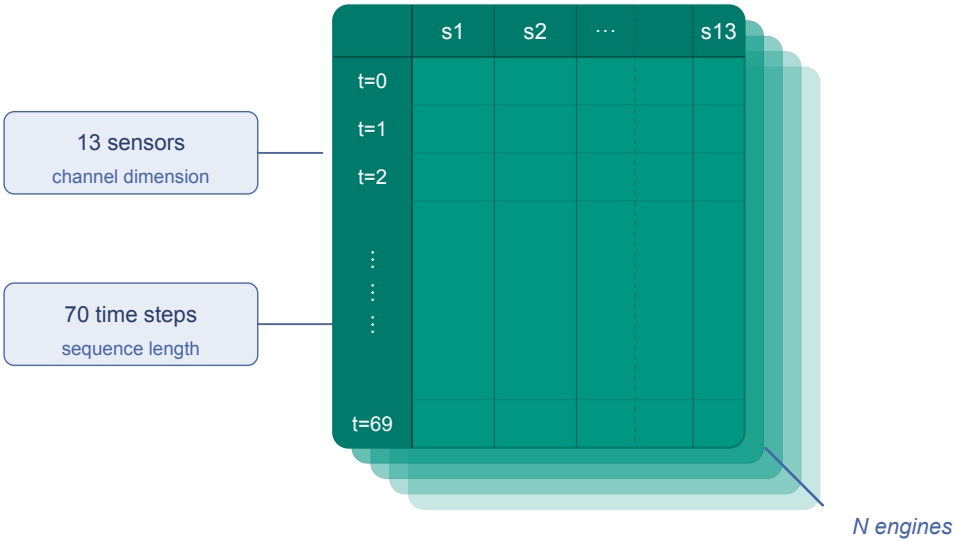

Figure 4.1: Structure of the input tensor for a single sliding window sample, consisting of 13 sensor channels over 70 consecutive time steps. Each slice in the depth direction represents one engine unit.

associated with each window is the RUL value at the final time step of that window. The resulting input strcture is illustrated in Figure 4.1.

4.1.2 Target definition (RUL label)

The raw RUL for each engine trajectory is computed as the difference between the end-of-life cycle t_{EOL} and the current cycle t:

$$\text{RUL}_{\text{raw}}(t) = t_{\text{EOL}} - t \tag{4.1}$$

A purely linear RUL label is known to dominate model training with the healthy regime, where no meaningful degradation signal is present. To address this, a piecewise linear transformatino is applied using a maximum RUL threshold $R_c = 125$ cycles. Early-life cycles, where the tru RUL exceeds R_c, are assigned a constant label of R_c, under the assumptioin that the engine shows no detectable degradation in this regime. Only once the trajectory enters the degradation region does the label decrease linearly toward zero at failure:

$$\text{RUL}(t) = min(\text{RUL}_{\text{raw}}(t), R_c). \tag{4.2}$$

This transformation prevents the model from being overwhelmed by healthy-state samples and focuses learning on the degradation region, which carries the operationally relevant signal.

Following the piecewise transfromation, RUL labels are normalized to [-1, 1] using a min-max scaling based on the training set:

$$\hat{y} = 2 * \frac{\mathrm{RUL}(t) - \mathrm{RUL}_{\mathrm{min}}}{\mathrm{RUL}_{\mathrm{max}} - \mathrm{RUL}_{\mathrm{min}}} - 1, \tag{4.3}$$

where $\mathrm{RUL}_{\mathrm{min}}$ and $\mathrm{RUL}_{\mathrm{max}} = R_c = 125$ by construction, so that a healthy engine at the cap maps to +1 and an engine at failure maps to -1. The same scaling parameters are applied to the test set without re-fitting.

4.2 Model Setup

All model variants used in this thesis share a common architecture base on a Domain-Adversarial Neural Netwrok (DANN) . The DANN consists of three components: a convolutional feature extractor, a label predictor, and an optional domain classifier connected via a Gradient Reversal Layer (GRL). Since the primary purpose of the models here is to serve as a stable substrate for feature importance analysis rather than to maximise transfer performance, the domain classifier branch is used selectively and the architecture is held fixed across all configurations.

The **Feature Extractor** processes the input tensor of shape $(Nx13xT)$ through four successive one-dimensional convolutional layers. The number of filters follows a halving schedule: 128, 64, 32, and 16 channels respectively, each with kernel size 5 and same-length padding of 2. After each convolutional layer, a LeakyReLu activation with negative slope 0.1 is applied. The temporal dimensions is collapsed using a global average pooling operation (AdaptiveAvgPool1d), which produces a fixed-length representation of 16 values regardless of input sequence length. A final fully connected layer projects this to a feature vector of length 64. Note that the slab height in Figure 4.2 is a visual convention representing the reducing channell count and does not imply a reduction in the temporal dimesnion, which remains 70 throughout.

The **Label Predictor** maps the 64-dimensioinal feature vector to a scalar RUL estimate through two fully connected layers: Linear $(64 \to 32)$ and Linear $(32 \to 1)$, each preceded by a ReLU activation. The output is a single value representing the predicted RUL.

4.3 Training Procedure

All models are trained to minimise the mean squared error (MSE) between the predicted and true normalised RUL values:

$$\mathscr{L} = \frac{1}{N} \sum_{i=1}^{N} (\hat{y}^{(i)} - y^{(i)})^2. \tag{4.4}$$

The models are trained with the following subset combinations.

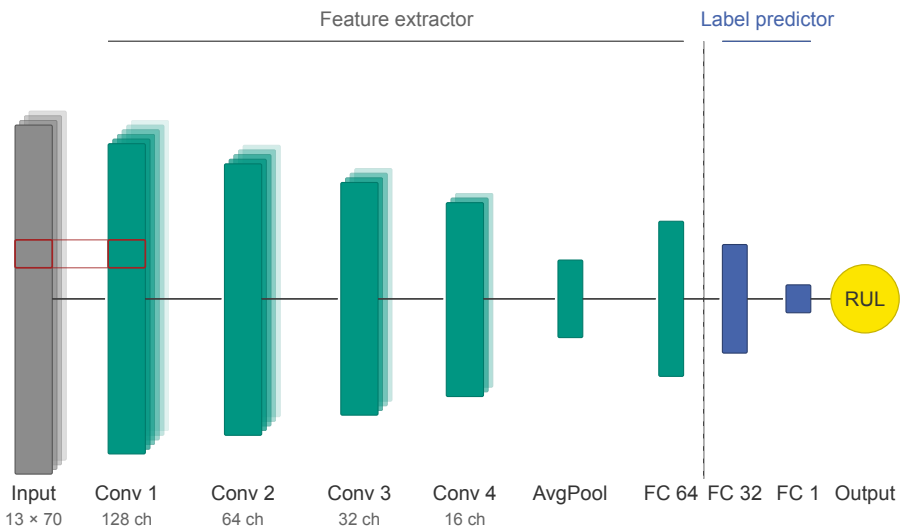

Figure 4.2: Architecture of the 1D convolutional neural network. The red window illustrates the receptive field of the convolutional kernel ($k = 5$). Green: feature extractor, blue: label predictor.

4.3.1 Model Configurations

Three training configurations are studied, differing only in the subset of CMAPSS data used during training. The four single-dataset models each train on one sub-dataset in isolation and serve as the baseline for feature importance behaviour under a single operating condition. The

six pairwise models each train on the concatenated data from two sub-datasets, allowing examination of how feature importance shifts when a model must generalise across two distinct operating regimes simultaneously. The full-dataset model trains on all four sub-datasets combined and represents the most demanding generalisation scenario. Table 4.2 gives an overview of all eleven model configurations.

Table 4.2: Overview of all model configurations used in this work.

Configuration	**Training data**	**No. of models**
Single	FD001	
	FD002	
	FD003	
	FD004	4
Pairwise	FD001 + FD002	
	FD001 + FD003	
	FD001 + FD004	
	FD002 + FD003	
	FD002 + FD004	
	FD003 + FD004	6
Full	FD001 + FD002 + FD003 + FD004	1
Total		**11**

4.4 Implementation of Feature Importance Methods

All eight feature importance methods are applied post-hoc to the trained models, without any modification of the learned weights. A central implementation challenge shared across all methods is the multivariate time-series structure of the input: each sample is a tensor of shape (N x 13 x 70), where the 13 sensor channels and 70 time steps together constitute the model's input spcae. Most standard feature importance implementations assume a tabular input where each column corresponds to an independent scalar feature. Applying these methods to a sequence model therefore requires deliberate desing dicisions about how features are defined, how perturbations are applied and how attributions over the temporal dimesnion are reduced to a per-sensor importance score. Unless stated otherwise, the full temporal extent of a sensor channel is treated as a single unit, perturbations are applied uniformly across all 70 time steps of a given sensor simultaneously and the resulting attributions are aggregated over time to yield a scalar importance score per sensor. This choice reflects the physically motivated view that a sensor's contribution to the RUL prediction should be attributed to the sensor as a whole rather than decomposed into individual time steps, which would result in a 910-dimensional attribution space that is difficutlt to interpret and compare across methods.

Individual Conditional Expectation (ICE): For each of the 13 sensors, a uniform grid of 20 values is constructed between the 5th and 95th empirical percentile of that sensor's observed values across the dataset. The percentile bound are used in place of the global minimum and maximum to avoid evaluatiing the model in the tails of the distribution, where training data is sparse and prediction may be unreliable. For each grid value g and each instance in the dataset, the entire time series of sensor j is replaced by the constant value g, that is, $x_{j,t} = g$ for all $t \in [1, 70]$, while all remaining sensor channels are kept at their observed values. This produces one ICE curve per instance, tracing how the predicted RUL changes as sensor j is swept across its range in isolation. The PDP is then computed as the pointwise mean of all ICE curves. Feature importance is derived as ICE-FI$_j$ (see 3.4), measuring the mean variance of ICE curves across all instances.

Feature Ablation: The implementation follows the grouped ablatioin scheme provided by Captums's *FeatureAblation*, in which all 70 timesteps belonging to a given sensor channel are masked simultaneously as a single group rather than independently. This sensor-grouped formulation is essential for the present setting: ablating individual timesteps in isolation would produce attributions across 910 independent input positions, making cross-method comparison difficult and reducing the physical interpretability of the results. When sensor j is ablated, its entire time series is replaced by the training set mean computed independently for each channel, representing the expected signal in the absence of informative variation. Attributions are defined as the absolute change in prediction relative to the unpertrubed output, and global feature importance is obtained by averaging these absolute differences over all instances in the evaluation set.

Shapley Additive Explanations: To maintain consistency with the grouped ablation scheme described above, the 13 sensor channels are treated as 13 players in the cooperative game rather than operating on the 910-dimensional flattened input. Exact Shapley values are computed by enumerating all $2^{13} = 8192$ possible sensor coalitions and evaluating the model on each, which is computatinoally tractable for this group size. When a sensor group is absent from a coalition, its timesteps are replaced by a mean valued baseline as in the ablation method. For each coalitioin S, the masked input is constructed by replacing the features of all sensors not in S with the baseline and the model is queried to obtain $v(S)$. The Shapley values are then computed from these chached coalition outputs, yielding a per-sensor attribution for each instance in the dataset. A visualistation of the Shapley values is provided in the results chapter and implemented using the *shap* library.

Local Interpretable Model-Agnostic Explanations: The implementation uses Captum's `LIME` with a Lasso linear surrogate as the interpretable model class. Since LIME operates by perturbing the input in an interpretable feature space and fitting a weighted linear regression, the (13 x 70) input tensor is first flattened to a 910-dimensional vector. For each instance to be explained, 4000 perturbed samples are drawn in the neighbourhood of the input, each evaluated by the model. A proximity-weighted Lasso regression($\alpha = 0.01$) is then iftted to approximate the model locally, with the regularisation term enforcing sparsity in the resulting attribution vector. The zero vector is used as the reference baseline. After fitting, the 910-dimesninal weight vector is reshaped back to (13 x 70) and the time dimensioin is averaged to obtain sensor-level

importance scores. The Lasso regularisation parameter α was selected to balance sparsity with attribution stability across instances.

Counterfactual Explanations: Counterfactual explanations are generated using the `DiCE` library, which implements a gradient -based search for minmally modified inputs that shift the model's prediction into a specified target range. As `DiCE` requires a tabular input representation, the (13 x 70) tensor is flattened to a 910-dimensional feature vector before being passed to the `DiCE` data interface, which is initialised using the training set to define plausible feature ranges. For each query instance, the desired output range is specified as a window around the original prediction, so that the counterfactual represents a plasuible alternative trajectory with a meaningfully differen predicted RUL. The distance between the original and counterfactual input is pnealised using an L_1 norm, which encourages sparse modifications, that is, changes concentrated in a small number of features. Feature importance is subseqently derived by computing the magnitued of change per sensor, aggregated across the 70 timesteps, which identifies which sensors were most frequently or substantially modified to achieve the target prediction shift.

Permutation Feature Importance: For each sensor channel, its values are permuted randomly across all N samples in the evaluation set, while all other channels remain at their original values. This destroys the statistical association between that sensor and the target RUL without altering the marginal distribution of the sensor's values, so any resulting drop in predictive performance can be attributed to the loss of that sensor's information. Performance is measured as MSE on the normaliised RUL scale. To account for variablity in the permutation, the procedure is repeated with 5 independent random permutations and the mean and standard deviation of the MSE increase are reported per sensor. The use of 5 repeats provides a sufficiently stable estimate for 13 sensors while keeping the total number of model evaluation manageable.

Accumulated Local Effects: The domain of each sensor is partitioned into 20 bins using quantile-based bin edges, so that each bin contains approximately the same number of samples. Quantile-based binning is preferred over unifrom binning because sensor distributions may be skewed after operating-condition-specific normalisation and equal-frequency bins ensure that local effect estimates are based on suffiecient data throughout the feature range. Bin membership for each sample is determined by the mean value of that sensor's time series, which reduces the (N x L) sensor values to a scalar per sample for assignment purposes. Within each bin, the local prediction difference is computed by replacing the entire time series of senso j with the lower and uper bin boundary values in turm, and recording the resulting change in model output. These local differences are averaged within each bin and then accumulated across bins following 3.16. The accumulated effect curve is centered to zero mean over the empirical distribution of sensor values. A scalar importance scoe is derived as the range of the centered ALE curve, reflecting the total span of predicted RUL variation attributable to that senos under the local conditional distribution.

Integrated Gradients: The implementation uses Captum's `IntegratedGradients` with 100 integration steps and the Gauss–Legendre quadrature approximation, which provides a more accurate estimate of the path integral than a simple Riemann sum for the same number of steps. The `multiply_by_inputs` flag is set to `True`, which scales each gradient by

the difference between the input and the baseline. This ensures that the completeness property holds, i.e. the sum of attributions over all input dimensions equals the difference between the model's prediction at the input and at the baseline. A zero-valued tensor of shape $(1 \times 13 \times 70)$ is used as the baseline, consistent with the choice made for SHAP and Feature Ablation. Since all sensor values lie in $[0,1]$ after MinMax normalisation, a zero baseline represents a meaningful reference state corresponding to the minimum observed sensor value in each channel. Attributions are computed per instance over the full (13×70) input space, absolute values are taken to reflect attribution magnitude regardless of sign, and the temporal dimension is averaged to yield a per-sensor importance score of shape $(13,)$.

4.5 Unified Feature Importance Representation

Because the eight feature importance methods produce attributions in different units and on different scales, a unified representation is required before any cross method comparison can be made. For each method, the raw feature-level attributions are reduced to a single importance score per sensro by aggregating over the temporal dimension where applicable. The resulting scores are then normalised using L_1 normalisation:

$$\mathrm{FI}_{\mathrm{L}_1} = \frac{|\mathrm{FI}_j|}{\sum_{k=1}^{13} |\mathrm{FI}_k|}, \tag{4.5}$$

so that the importance scores for all features sum to one for any given instance or aggregate. This normaliation ensures that the magnitude of attributions is comparable across methods regardless of their original scale, enabling direct visual and quantitative comparison in the results chapter.

5 Results

5.1 Verification of FI algorithms

section title: Algorithmic Verification of Feature Importance Methods

keep it general Feature importance algo & model

5.2 Comparison: Single vs Combined Models

global comparison

sensor comparison

time comparison

5.3 Feature Importance Results per Method

per method analysis

5.4 Comparison

correlation matrices feature importance of all methods, summed together -> heat maps

time component

5.5 Discussion

6 Conclusion and Outlook

6.1 Summary of Contributions

6.2 Answering the Research Questions

6.3 Outlook

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Appendix

A.1 Anhang 1

Table A.1: Simulationsergebnisse, die den Lesefluss stören.

	$w = 8$			$w = 16$		
	$t = 0$	$t = 1$	$t = 2$	$t = 0$	$t = 1$	$t = 2$
$dir = 1$						
c	0,0790	0,1692	0,2945	0,3670	0,7187	3,1815
c	-0,8651	50,0476	5,9384	-9,0714	297,0923	46,2143
c	124,2756	-50,9612	-14,2721	128,2265	-630,5455	-381,0930
$dir = 0$						
c	0,0357	1,2473	0,2119	0,3593	-0,2755	2,1764
c	-17,9048	-37,1111	8,8591	-30,7381	-9,5952	-3,0000
c	105,5518	232,1160	-94,7351	100,2497	141,2778	-259,7326

A.2 Anhang 2

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